

In23-S4-IN2601 - Computer Graphics

Hardware components of computer graphics >

By :

W.A.G.Nilnuwani



Learning Objectives

- Understand the main components of computer graphics hardware
- Explain how different display devices work
- Compare raster, vector, and flat-panel display technologies
- Describe graphics input, output, and hard-copy devices
- Understand graphics workstations, networks, and Internet-based graphics
- Perform basic calculations related to graphics hardware

Refresh Cathode-Ray Tubes

- A **CRT** is an old type of display device used in televisions and computer monitors.
- It works by using an **electron beam** to create images on the screen.
- **How it works**
 - ❑ Inside the CRT, there is an **electron gun**.
 - ❑ The electron gun shoots a beam of electrons toward the screen.
 - ❑ The screen is coated with a material called **phosphor**.
 - ❑ When the electron beam hits the phosphor, it glows and produces a **small spot of light**.
 - ❑ By moving the beam to many positions on the screen, a full image is formed.

Refresh Cathode-Ray Tubes

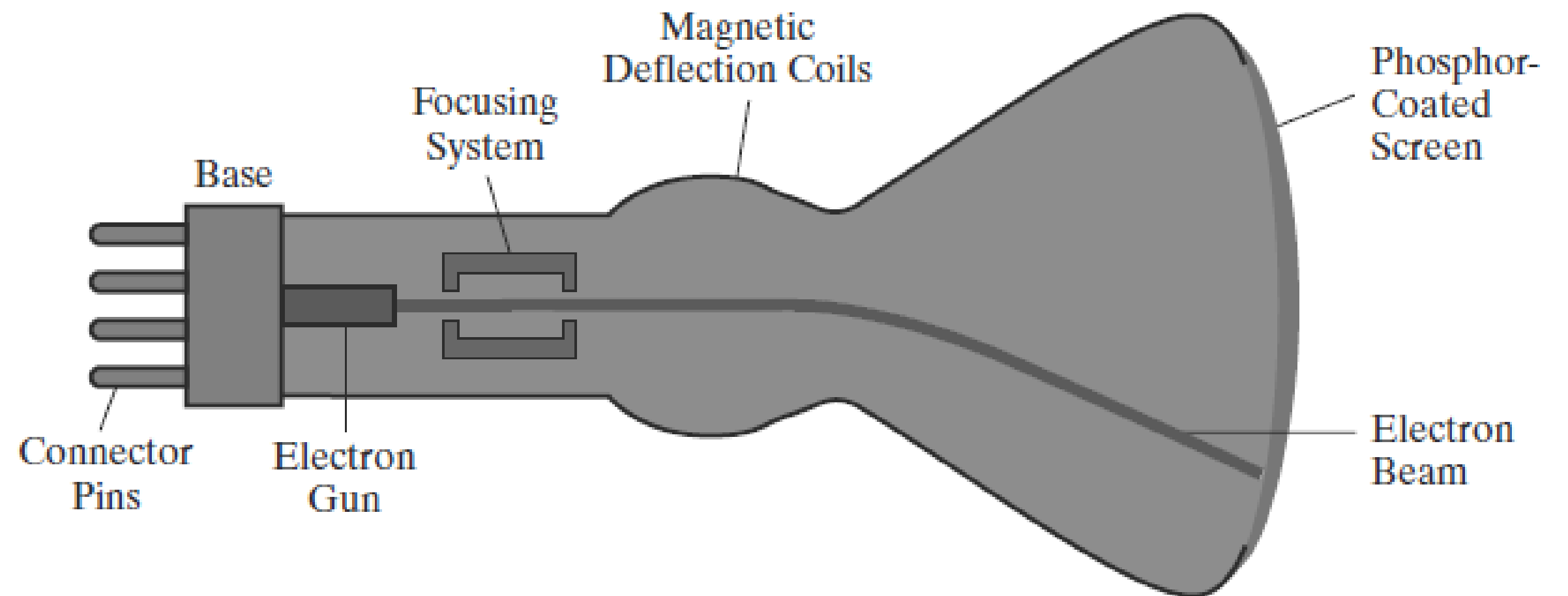


FIGURE 1
Basic design of a magnetic-deflection CRT.

Refresh Cathode-Ray Tubes

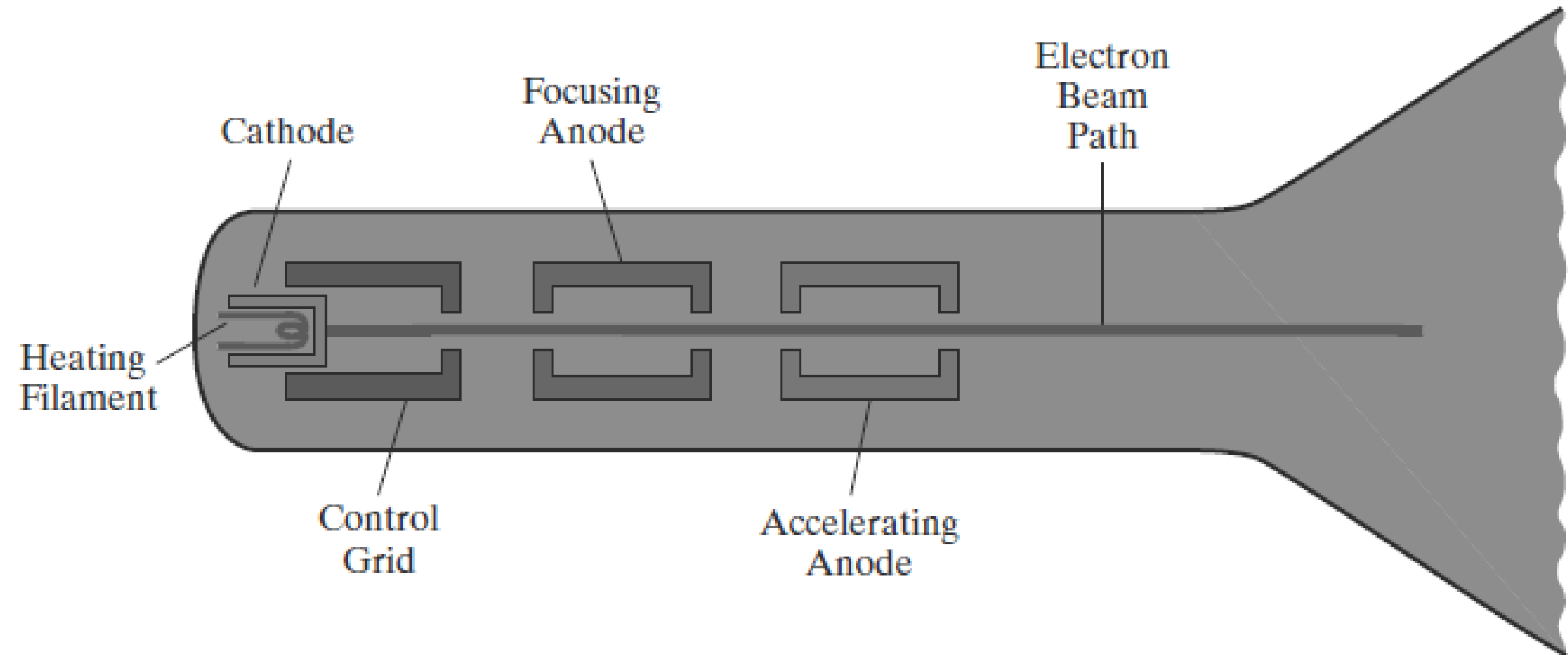


FIGURE 2

Operation of an electron gun with an accelerating anode.

Refresh Cathode-Ray Tubes

- Why “refresh” is needed?
 - ❑ The light produced by phosphor does **not stay for long**.
It fades very quickly.
 - ❑ So, to keep the picture visible, the electron beam must draw the same image **again and again very fast**.
 - ❑ This repeated redrawing is called **refreshing**.
 - ❑ The number of times the picture is redrawn in one second is called the **refresh rate**.
- For example:
 - ❑ **60 Hz** means the screen is refreshed **60 times per second**
 - ❑ If the refresh rate is too low, the screen may appear to **flicker**.

Refresh Cathode-Ray Tubes

- **How is brightness controlled?**

- ❑ The **control grid** controls how many electrons are allowed to pass.

- ❑ If the control grid has a **high negative voltage**, it repels electrons strongly and can **stop the beam completely**.

- ❑ If the grid has a **smaller negative voltage**, some electrons can pass through.

- So

- ❑ **More electrons reaching the screen** - brighter spot

- ❑ **Fewer electrons reaching the screen** - dimmer spot

- That means the **brightness (intensity)** of each point on the screen is controlled by the **control grid voltage**.

Refresh Cathode-Ray Tubes

- **Why does the light fade quickly?**
 - The phosphor does not glow forever.
 - Once the excited electrons return to their normal state, the light starts to disappear.
 - So the glowing spot **fades quickly**.
 - That is why the picture must be **refreshed again and again**.
- **What decides the color of the light?**
 - The **color** of the light depends on the energy difference between:
 - the excited state
 - the ground state
 - Different phosphor materials produce different colors.

Refresh Cathode-Ray Tubes

What is resolution?

- Resolution means the maximum number of points (pixels/spots) that can be shown clearly on the screen without overlapping.
- Resolution tells us how much detail a screen can display.
- A higher resolution means more points on the screen, sharper image and more detail.
- Resolution can be described in two ways.
- **Common way** is number of points across the screen horizontally and vertically
- for example **1280 × 1024** (1280 points across and 1024 points down)
- **More accurate way** is number of points that can be displayed **per centimetre** which shows how closely points can be packed together.

Refresh Cathode-Ray Tubes

How does intensity distribute in a light spot?



FIGURE 4

Intensity distribution of an illuminated phosphor spot on a CRT screen.



FIGURE 5

Two illuminated phosphor spots are distinguishable when their separation is greater than the diameter at which a spot intensity has fallen to 60 percent of maximum.

Raster-Scan Displays

- The most common type of graphics monitor employing a CRT is the **raster-scan display**, based on television technology.
- In a raster-scan system, the electron beam is swept across the screen, one row at a time, from top to bottom. Each row is referred to as a **scan line**.
- Picture definition is stored in a memory area called the **refresh buffer** or **frame buffer**, where the term **frame** refers to the total screen area.
- Each screen spot that can be illuminated by the electron beam is referred to as a **pixel** or **pel** (shortened forms of picture element).
- it is also sometimes called a **color buffer**
- Another property of video monitors is **aspect ratio**, which is now often defined as the number of pixel columns divided by the number of scan lines that can be displayed by the system

$$\text{Aspect ratio} = \frac{\text{number of pixel columns}}{\text{number of scan lines}}$$

Raster-Scan Displays

Determine the resolution (pixels per centimeter) in the x and y directions for the video monitor in use on your system. Determine the aspect ratio, and explain how relative proportions of objects can be maintained on your system.

Raster-Scan Displays

Assume a monitor has, Resolution = **1920 × 1080** Width = **53 cm** Height = **30 cm**

In X direction,

$$\text{Pixels per cm in x} = \frac{\text{Number of horizontal pixels}}{\text{Screen width in cm}} \quad 1920/53 = 36.23 \text{ pixels/cm}$$

In y direction,

$$\text{Pixels per cm in y} = \frac{\text{Number of vertical pixels}}{\text{Screen width in cm}} \quad 1080/30 = 36 \text{ pixels/cm}$$

$$\text{Aspect ratio} = \frac{\text{Number of horizontal pixels}}{\text{Number of Vertical pixels}} \quad 1920/1080 = 16/9$$

Raster-Scan Displays

- The range of colors or shades of gray that can be displayed on a raster system depends on both the types of phosphor used in the CRT and the number of bits per pixel available in the frame buffer.
 - 1 bit, 24 bits
- The number of bits per pixel in a frame buffer is sometimes referred to as either the **depth** of the buffer area or the number of **bit planes**
- For example, a system with 24 bits per pixel and a screen resolution of 1024 by 1024 requires, 3 MB the storage required for the refresh buffer

Raster-Scan Displays

- **Consider three different raster systems with resolutions of 800 by 600, 1280 by 960, and 1680 by 1050. What size frame buffer (in bytes) is needed for each of these systems to store 16 bits per pixel?**
- **How much storage is required for each system if 32 bits per pixel are to be stored?**

Raster-Scan Displays

Frame buffer size (bytes) = Resolution × bits per pixel ÷ 8

For 800 × 600,

Total pixels - $800 \times 600 = 480,000$

For 16 bits per pixel - $480,000 \times 16 = 7,680,000$ bits

$= 7,680,000 / 8 = 960,000$ bytes

For 32 bits per pixel - $480,000 \times 32 = 15,360,000$ bits

- $15,360,000 / 8 = 1,920,000$ bytes

Raster-Scan Displays

- As each screen refresh takes place, we tend to see each frame as a smooth continuation of the patterns in the previous frame, so long as the refresh rate is not too low. Below about 24 frames per second, we can usually perceive a gap between successive screen images, and the picture appears to flicker.
- Sometimes, refresh rates are described in units of cycles per second, or hertz (Hz), where a cycle corresponds to one frame. Using these units, we would describe a refresh rate of 60 frames per second as simply 60 Hz

Random-Scan Displays

- Search and write the operating features of Random-Scan Displays.

Color CRT Monitors

- A CRT monitor displays color pictures by using a combination of phosphors that emit different-colored light.
- One way to display color pictures is to coat the screen with layers of different colored phosphors. The emitted color depends on how far the electron beam penetrates into the phosphor layers. This approach, called the **beam-penetration** method, typically used only two phosphor layers: red and green.
- **Shadow-mask** methods are commonly used in raster-scan systems (including color TV) because they produce a much wider range of colors than the beam penetration method. This approach is based on the way that we seem to perceive colors as combinations of red, green, and blue components, called the **RGB color model**.

Color CRT Monitors

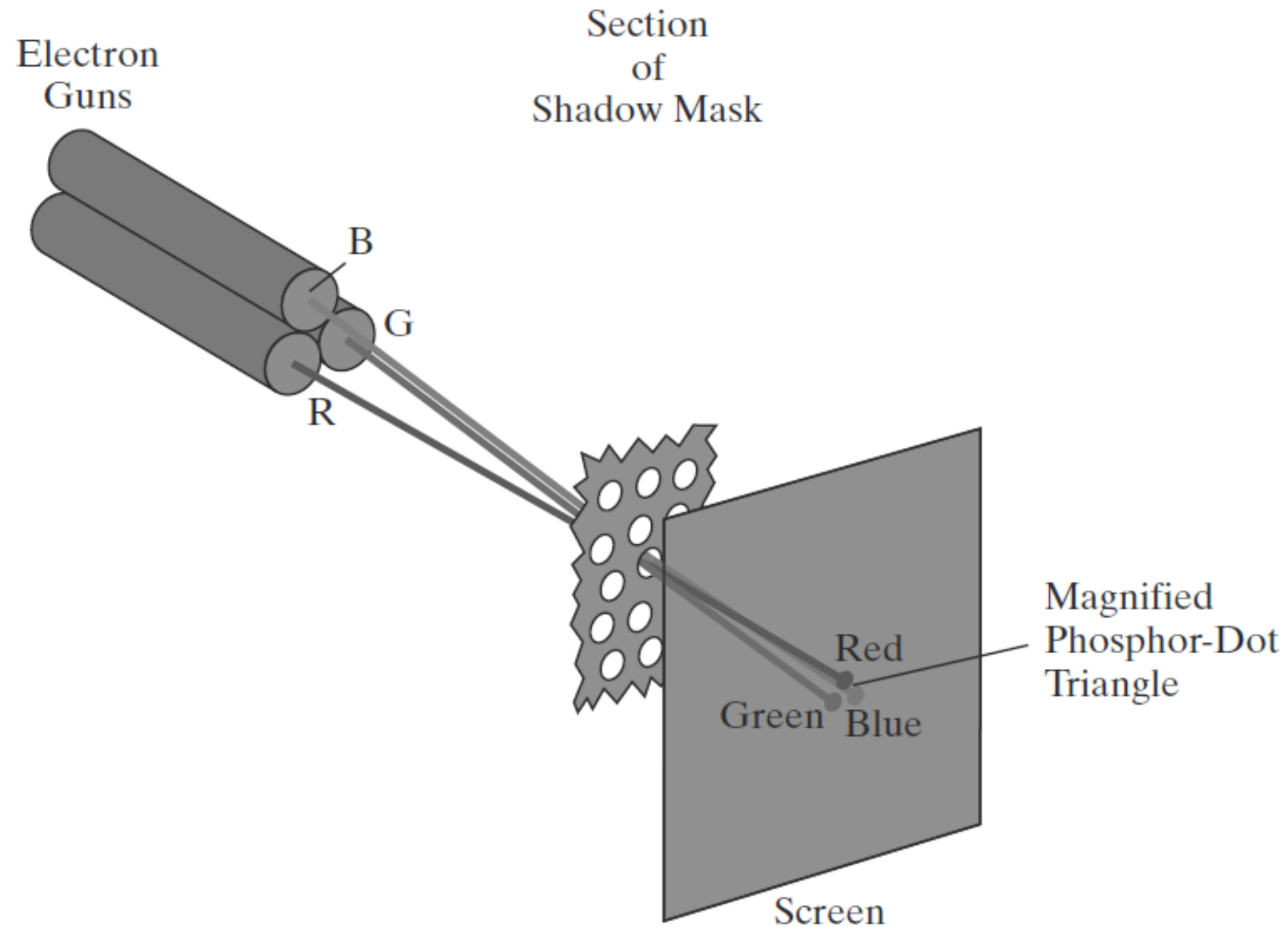


FIGURE 9
Operation of a delta-delta, shadow-mask CRT. Three electron guns, aligned with the triangular color-dot patterns on the screen, are directed to each dot triangle by a shadow mask.

Color CRT Monitors

- **How are color graphics systems used with CRT display devices in inexpensive home computers and video games, and why does using a color TV with an RF modulator reduce image quality?**

Color CRT Monitors

- **How are color graphics systems used with CRT display devices in inexpensive home computers and video games, and why does using a color TV with an RF modulator reduce image quality?**
- Some old color graphics systems displayed images on a normal color TV instead of using a special computer monitor.
- A CRT display device is the old bulky screen technology used in televisions and monitors. In low-cost home computers and video game systems, manufacturers often used a regular color TV as the display device because it was cheaper.
- But a TV cannot directly understand the computer's picture data. So the system used a device called an RF modulator.
- What the RF modulator does
- The RF modulator acts like a converter which takes
 - the color information
 - the brightness/intensity information

and combines them into a form that looks like a TV broadcast signal.

Color CRT Monitors

- **Composite monitors** are adaptations of TV sets that allow bypass of the broadcast circuitry. These display devices still require that the picture information be combined, but no carrier signal is needed.
- Color CRTs in graphics systems are designed as **RGB monitors**. These monitors use shadow-mask methods and take the intensity level for each electron gun (red, green, and blue) directly from the computer system without any intermediate processing. High-quality raster-graphics systems have 24 bits per pixel in the frame buffer, allowing 256 voltage settings for each electron gun and nearly 17 million color choices for each pixel. An RGB color system with 24 bits of storage per pixel is generally referred to as a **full-color system** or a **true-color system**.

Flat-Panel Displays

- The term flat-panel display refers to a class of video devices that have reduced volume, weight, and power requirements compared to a CRT.
- **What are emissive and non-emissive flat-panel displays? Explain with examples.**

Three-Dimensional Viewing Devices

- Graphics monitors for the display of three-dimensional scenes have been devised using a technique that reflects a CRT image from a vibrating, flexible mirror.
- As the varifocal mirror vibrates, it changes focal length. These vibrations are synchronized with the display of an object on a CRT so that each point on the object is reflected from the mirror into a spatial position corresponding to the distance of that point from a specified viewing location.

Three-Dimensional Viewing Devices

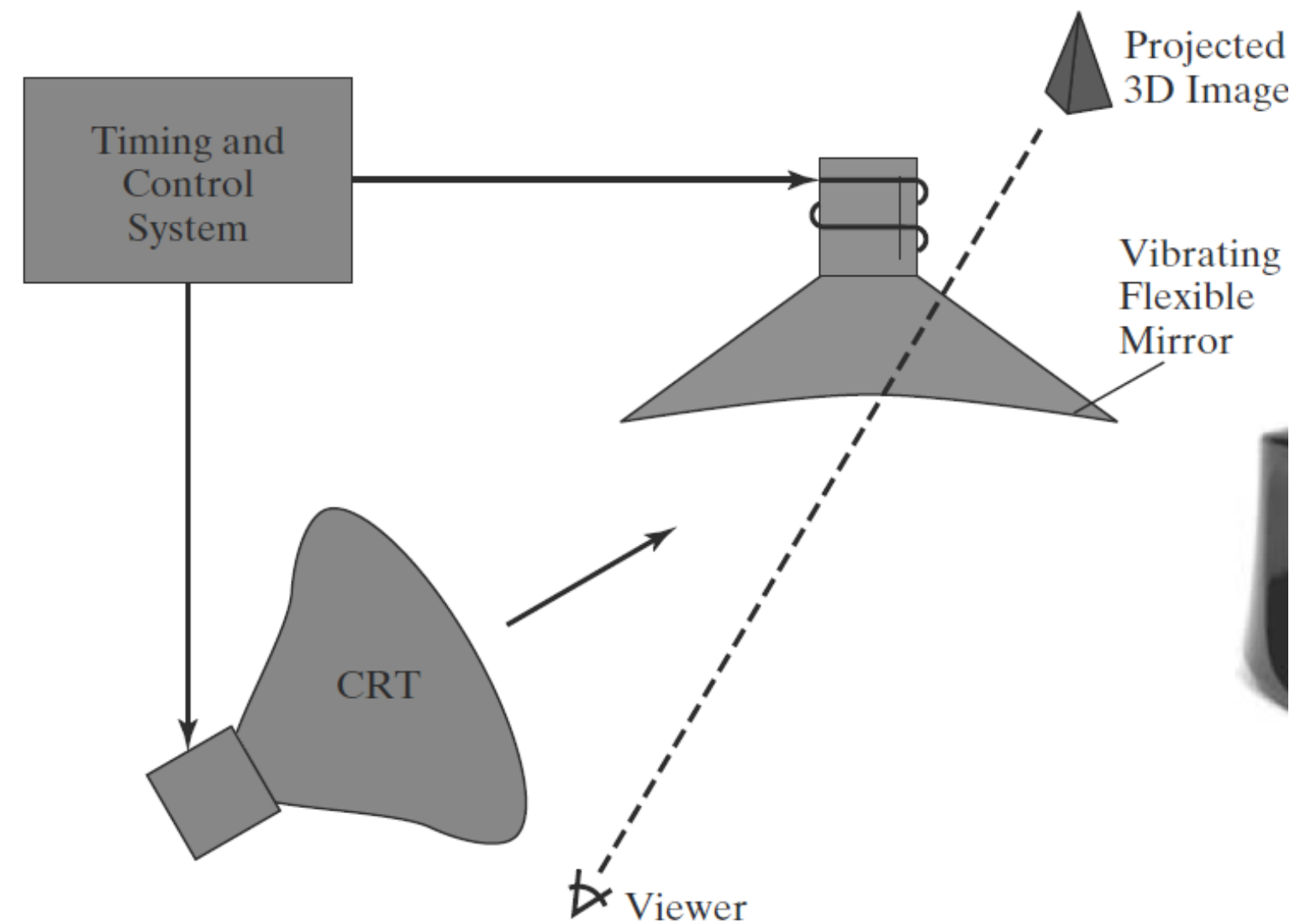


FIGURE 14
Operation of a three-dimensional display system using a vibrating mirror that changes focal length to match the depths of points in a scene.

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Stereoscopic and Virtual-Reality Systems

- Another technique for representing a three-dimensional object is to display stereoscopic views of the object. This method does not produce true three dimensional images, but it does provide a three-dimensional effect by presenting a different view to each eye of an observer so that scenes do appear to have depth.
- To obtain a stereoscopic projection, we must obtain two views of a scene generated with viewing directions along the lines from the position of each eye (left and right) to the scene. We can construct the two views as computer-generated scenes with different viewing positions, or we can use a stereo camera pair to photograph an object or scene. When we simultaneously look at the left view with the left eye and the right view with the right eye, the two views merge into a single image and we perceive a scene with depth.

Stereoscopic and Virtual-Reality Systems



FIGURE 15

Glasses for viewing a stereoscopic scene in 3D. (Courtesy of XPAND, X6D USA Inc.)

Stereoscopic and Virtual-Reality Systems

Describe the role of stereoscopic viewing in virtual-reality environments.

Stereoscopic and Virtual-Reality Systems

Describe the role of stereoscopic viewing in virtual-reality environments.

- Stereoscopic viewing is also a component in **virtual-reality** systems, where users can step into a scene and interact with the environment. A headset containing an optical system to generate the stereoscopic views can be used in conjunction with interactive input devices to locate and manipulate objects in the scene.
- A sensing system in the headset keeps track of the viewer's position, so that the front and back of objects can be seen as the viewer "walks through" and interacts with the display. Another method for creating a virtual-reality environment is to use projectors to generate a scene within an arrangement of walls, where a viewer interacts with a virtual display using stereoscopic glasses and data gloves.

Raster-Scan Systems

- Interactive raster-graphics systems typically employ several processing units. In addition to the central processing unit (CPU), a special-purpose processor, called the **video controller** or **display controller**, is used to control the operation of the display device.
- Here, the frame buffer can be anywhere in the system memory, and the video controller accesses the frame buffer to refresh the screen. In addition to the video controller, more sophisticated raster systems employ other processors as coprocessors and accelerators to implement various graphics operations.

Raster-Scan Systems

- Organization of a simple raster system is shown in Figure.

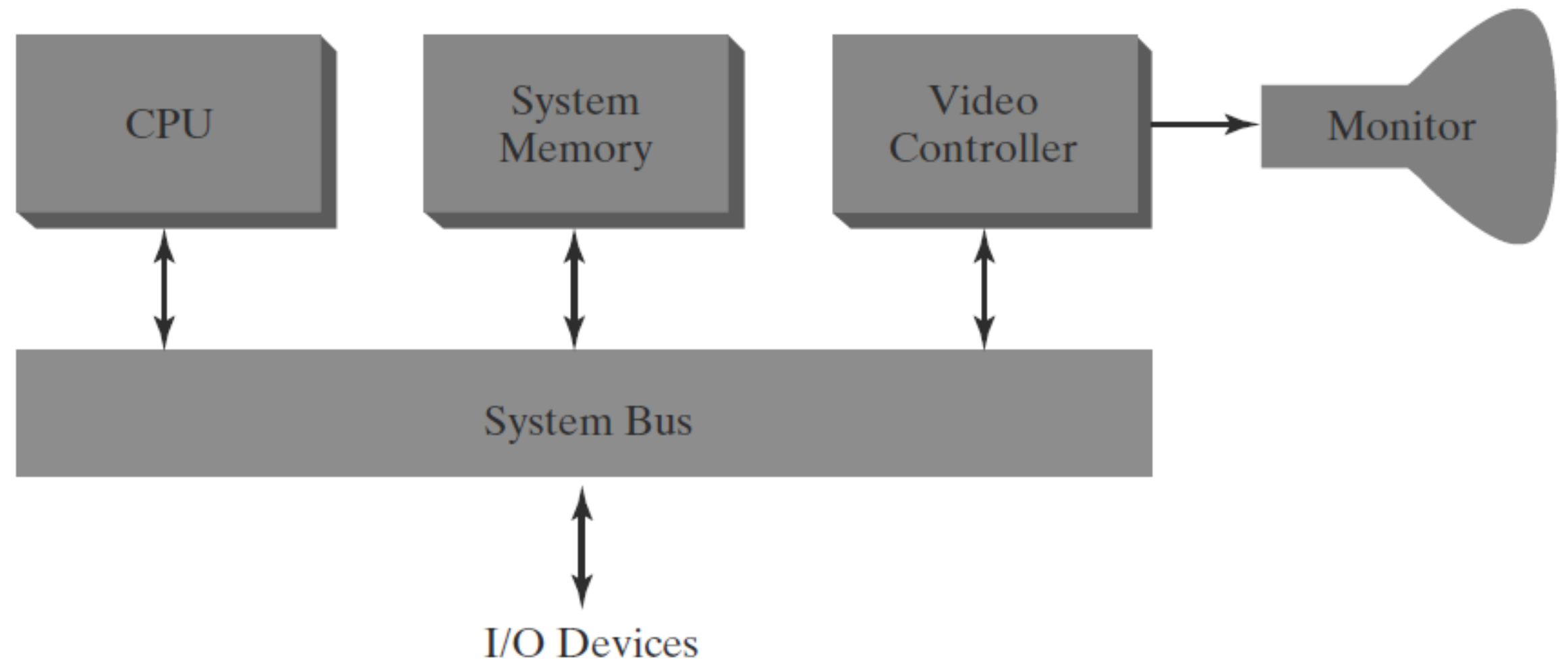


FIGURE 16

Architecture of a simple raster-graphics system.

Raster-Scan Systems

Is the raster-scan system architecture described in traditional computer graphics still used in modern graphics systems? Explain your answer.

Raster-Scan Systems

Is the raster-scan system architecture described in traditional computer graphics still used in modern graphics systems? Explain your answer.

The basic raster-scan concept is still used in modern graphics systems. Modern displays still refresh the screen by reading image data from a frame buffer and sending it to the screen in a raster pattern. However, the old simple video controller has now evolved into a much more advanced GPU and display engine, which handle rendering, acceleration, multiple displays, and other graphics functions more efficiently. So, the principle remains the same, but the hardware implementation is far more sophisticated today

Graphics Workstations and Viewing Systems

Today, most graphics monitors work as **raster-scan displays**. Both **CRT monitors** and **flat-panel displays** are used.

- A **graphics workstation** is a computer system designed to handle graphics tasks. These systems can range from:
 - ❑ Personal computers (640×480 , 1280×1024)
 - ❑ Graphics workstations (1280×1024 , 1600×1200)
 - ❑ High-definition graphics systems (2560×2048)
 - ❑ Large viewing systems (Multi-panel displays, Curved viewing screens)

Input Devices

- Graphics workstations can make use of various devices for data input. Most systems have a keyboard and one or more additional devices specifically designed for interactive input. These include a mouse, trackball, spaceball, and joystick. Some other input devices used in particular applications are digitizers, dials, button boxes, data gloves, touch panels, image scanners, and voice systems
- **Write a short note for each of the above input device listed.**

Hard-Copy Devices

- Hard-copy devices are used to produce permanent output of computer graphics in formats such as transparencies, slides, film, paper prints, and plotted drawings.
- The quality of the output depends mainly on **dot size** and **resolution** (dots per inch or lines per inch).
- These devices are broadly classified into **impact** and **nonimpact** printers.
- Impact printers, such as line printers and dot-matrix printers, create output by striking an inked ribbon against paper.
- Non-impact printers, including **laser**, **ink-jet**, **electrostatic**, and **electro-thermal** printers, produce output without direct mechanical contact.
- Color printing is achieved by combining color pigments, while plotters, especially **ink-jet and pen plotters**, are commonly used for drafting and detailed drawings.
- Hard-copy devices provide various methods for converting digital graphics into physical form.

Hard-Copy Devices



FIGURE 24

A picture generated on a dot-matrix printer, illustrating how the density of dot patterns can be varied to produce light and dark areas. (*Courtesy of Apple Computer, Inc.*)

Graphics Networks

- So far, we have mainly considered graphics applications on an isolated system with a single user. However, multiuser environments and computer networks are now common elements in many graphics applications. Various resources, such as processors, printers, plotters, and data files, can be distributed on a network and shared by multiple users.
- A graphics monitor on a network is generally referred to as a **graphics server**, or simply a **server**. Often, the monitor includes standard input devices such as a keyboard and a mouse or trackball. In that case, the system can provide input, as well as being an output server.
- The computer on the network that is executing a graphics application program is called the **client**, and the output of the program is displayed on a server.

Graphics Networks

- A workstation that includes processors, as well as a monitor and input devices, can function as both a server and a client.
- When operating on a network, a client computer transmits the instructions for displaying a picture to the monitor (server). Typically, this is accomplished by collecting the instructions into packets before transmission instead of sending the individual graphics instructions one at a time over the network. Thus, graphics software packages often contain commands that affect packet transmission, as well as the commands for creating pictures.

Graphics Networks

How long would it take to load an 800 by 600 frame buffer with 16 bits per pixel, if 10^5 bits can be transferred per second? How long would it take to load a 32-bit-per-pixel frame buffer with a resolution of 1680 by 1050 using this same transfer rate?

Graphics Networks

Time = Total number of bits / Transfer rate

transfer rate is 100,000 bits per second

1) 800 × 600 frame buffer with 16 bits per pixel

Total number of pixels = 800 × 600 = 480,000

Total number of bits = 480,000 × 16 = 7,680,000 bits

Time required = 7,680,000 / 100,000 = 76.8 seconds

2) 1680 × 1050, 32 bits/pixel - 564.48 seconds (about 9 min 24 s)

- **send packets efficiently**
- **reduce unnecessary data transfer**
- **sometimes send drawing commands instead of full image data**
- **use faster networks for high-quality graphics applications**

Graphics on the Internet

- A great deal of graphics development is now done on the global collection of computer networks known as the **Internet**. Computers on the Internet communicate using *transmission control protocol/internet protocol* (TCP/IP).
- The **World Wide Web** provides a hypertext system that allows users to locate and view documents that can contain text, graphics, and audio. Resources, such as graphics files, are identified by a *uniform (or universal) resource locator (URL)*. Each URL contains two parts: (1) the protocol for transferring the document, and (2) the server that contains the document and, optionally, the location (directory) on the server.
- For example, the URL *http://www.siggraph.org/* indicates a document that is to be transferred with the *hypertext transfer protocol (http)* and that the server is *www.siggraph.org*, which is the home page of the Special Interest Group in Graphics (SIGGRAPH) of the Association for Computing Machinery.
- Another common type of URL begins with *ftp://*. This identifies a site that accepts *file transfer protocol* (FTP) connections, through which programs or other files can be downloaded.

Graphics on the Internet

- Documents on the Internet can be constructed with the *Hypertext Markup Language (HTML)*. The development of HTML provided a simple method for describing a document containing text, graphics, and references (hyperlinks) to other documents.
- Although resources could be made available using HTML and URL addressing, it was difficult originally to find information on the Internet.
- Subsequently, the **National Center for Supercomputing Applications (NCSA)** developed a “browser” called **Mosaic**, which made it easier for users to search for Web resources. The Mosaic browser later evolved into the browser called **Netscape** Navigator. In turn, Netscape Navigator inspired the creation of the Mozilla family of browsers, whose most well-known member is, perhaps, **Firefox**.
- HTML provides a simple method for developing graphics on the Internet, but it has limited capabilities. Therefore, other languages have been developed for Internet graphics applications.